

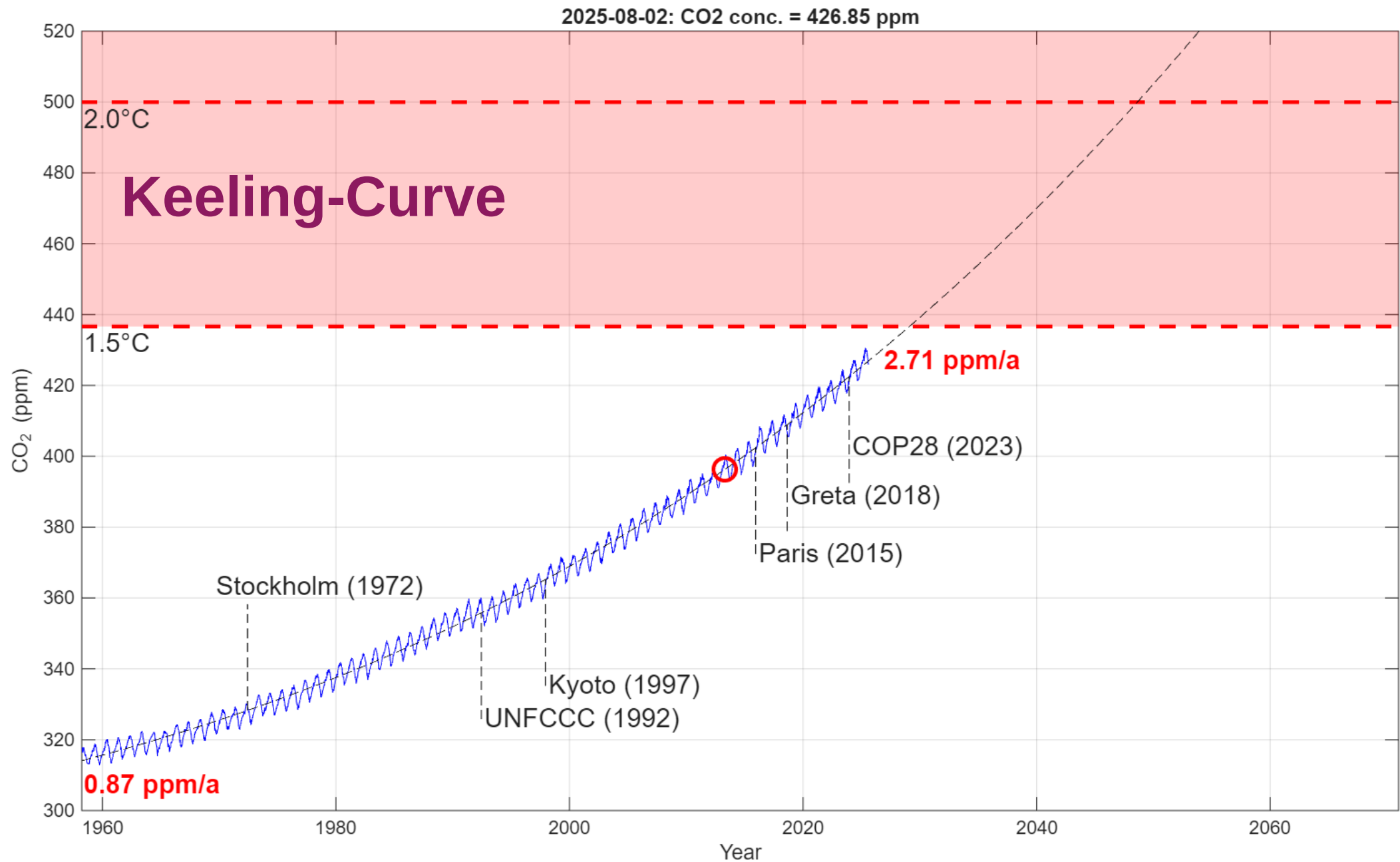
OST
Ostschweizer
Fachhochschule

Global Climate Compensation

Henrik Nordborg

2025-09-08

Renewable Energy and Environmental Technology



International climate politics has failed!

Is this a problem?

Why did we fail?

What do we do now?



The dangers of climate change

Die Klimakrise: Stehen wir am Abgrund oder sind wir schon einen Schritt weiter?

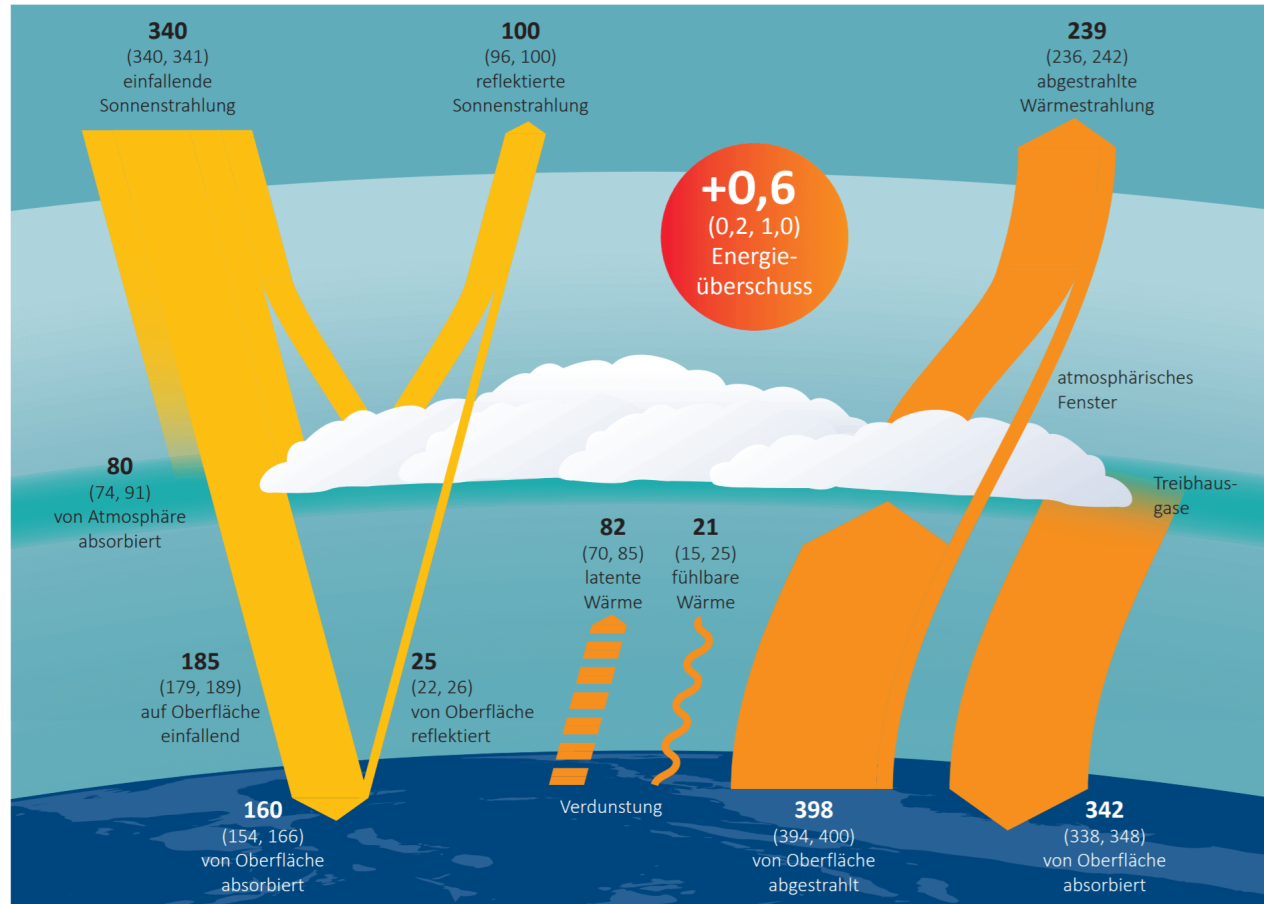
Henrik Nordborg

2025-09-08

The Physics of Climate Change

Der Klimawandel bringt die Energiebilanz der Erde aus dem Gleichgewicht

in Watt pro Quadratmeter

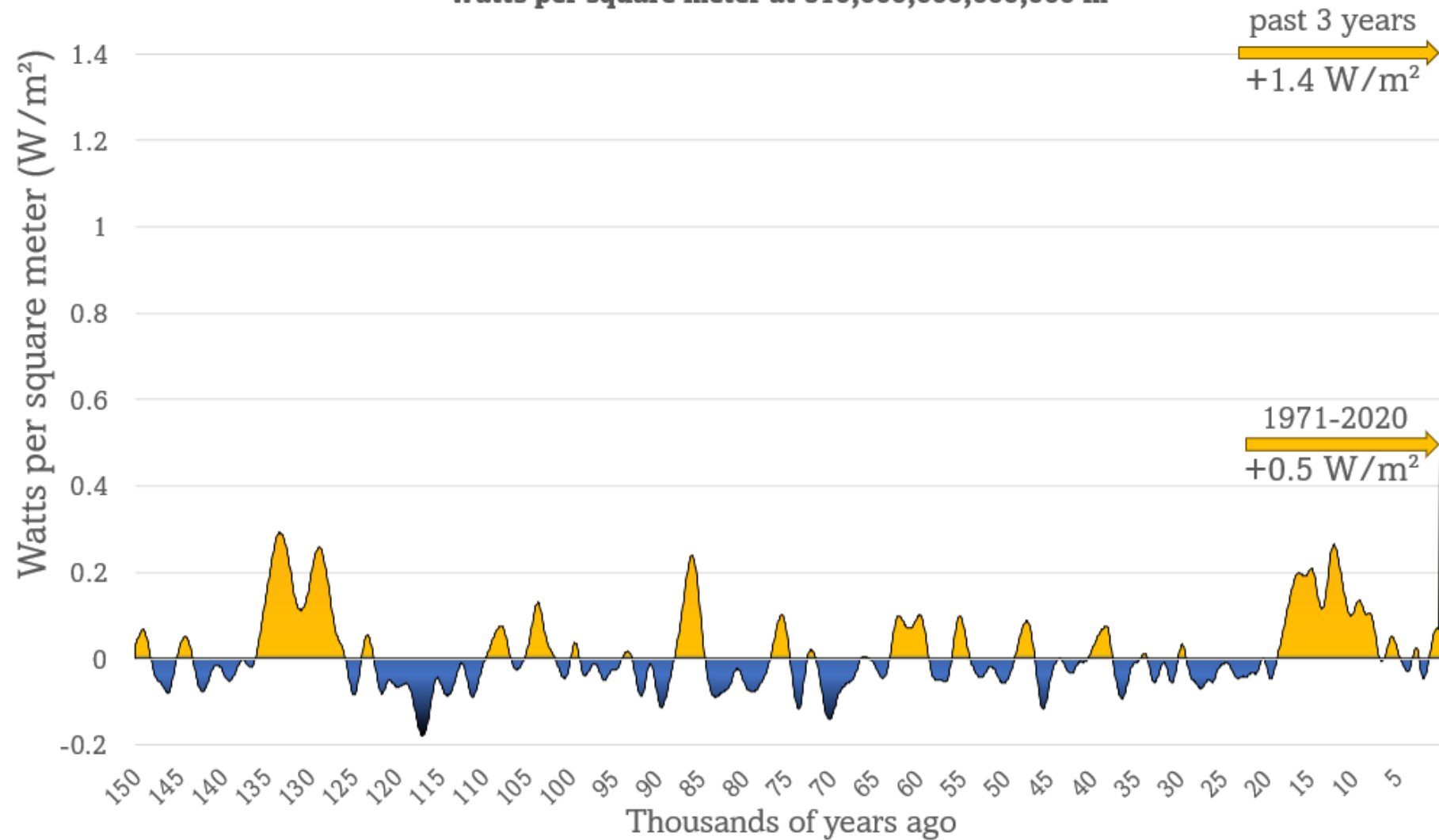


<https://correctiv.org/faktencheck/2022/10/27/doch-co2-hat-einen-einfluss-auf-den-klimawandel>

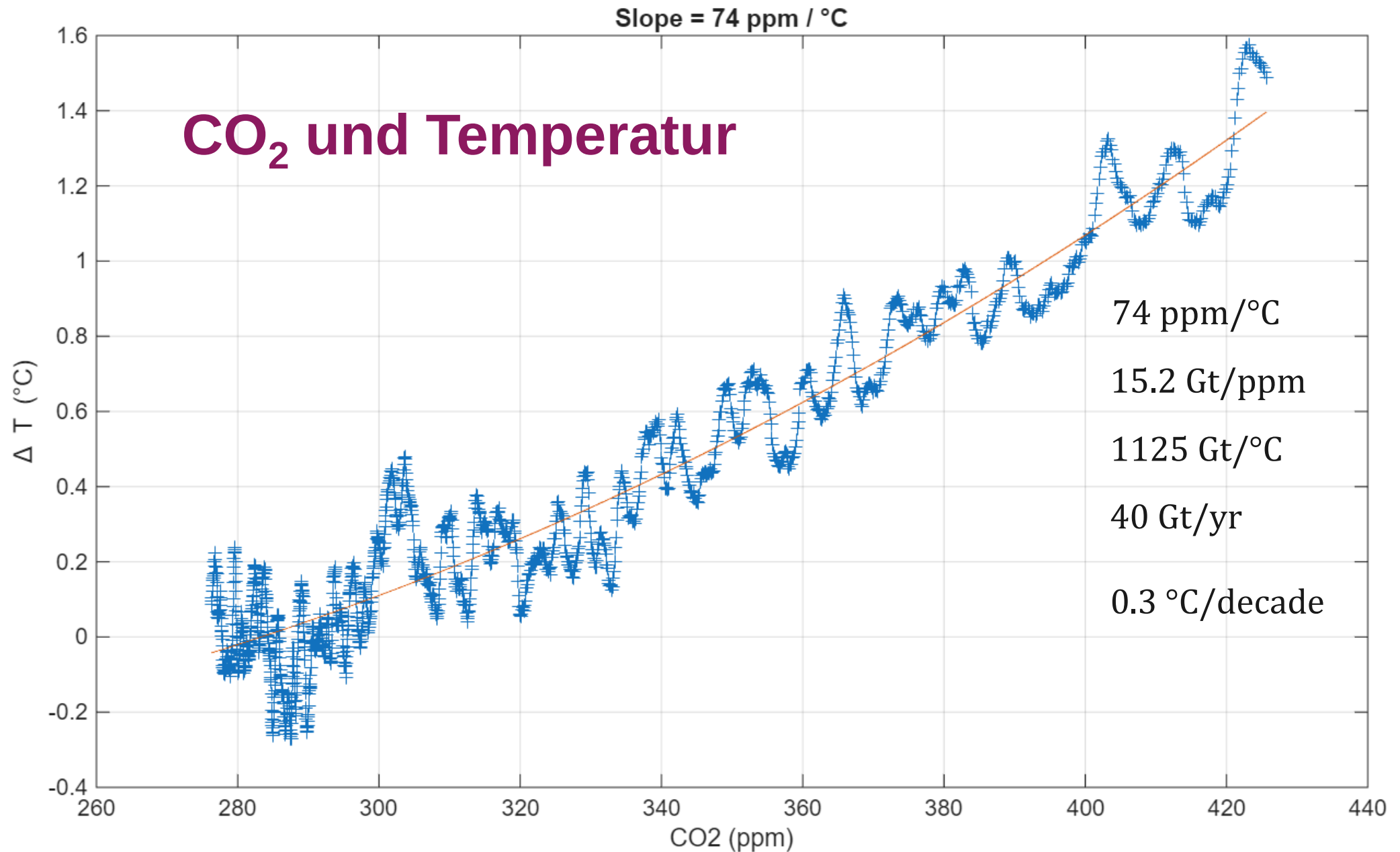
- The atmosphere allows visible (short-wave) light to pass through.
- The absorbed power heats the planet (160 W/m²)
- A total of around 82,000 TW, 4000 times human consumption.
- In order for the temperature of the earth to remain constant, the same amount of power must be emitted.
- However, the infrared (long-wave) heat radiation is partially absorbed by the greenhouse gases in the atmosphere.
- This causes the temperature to rise.

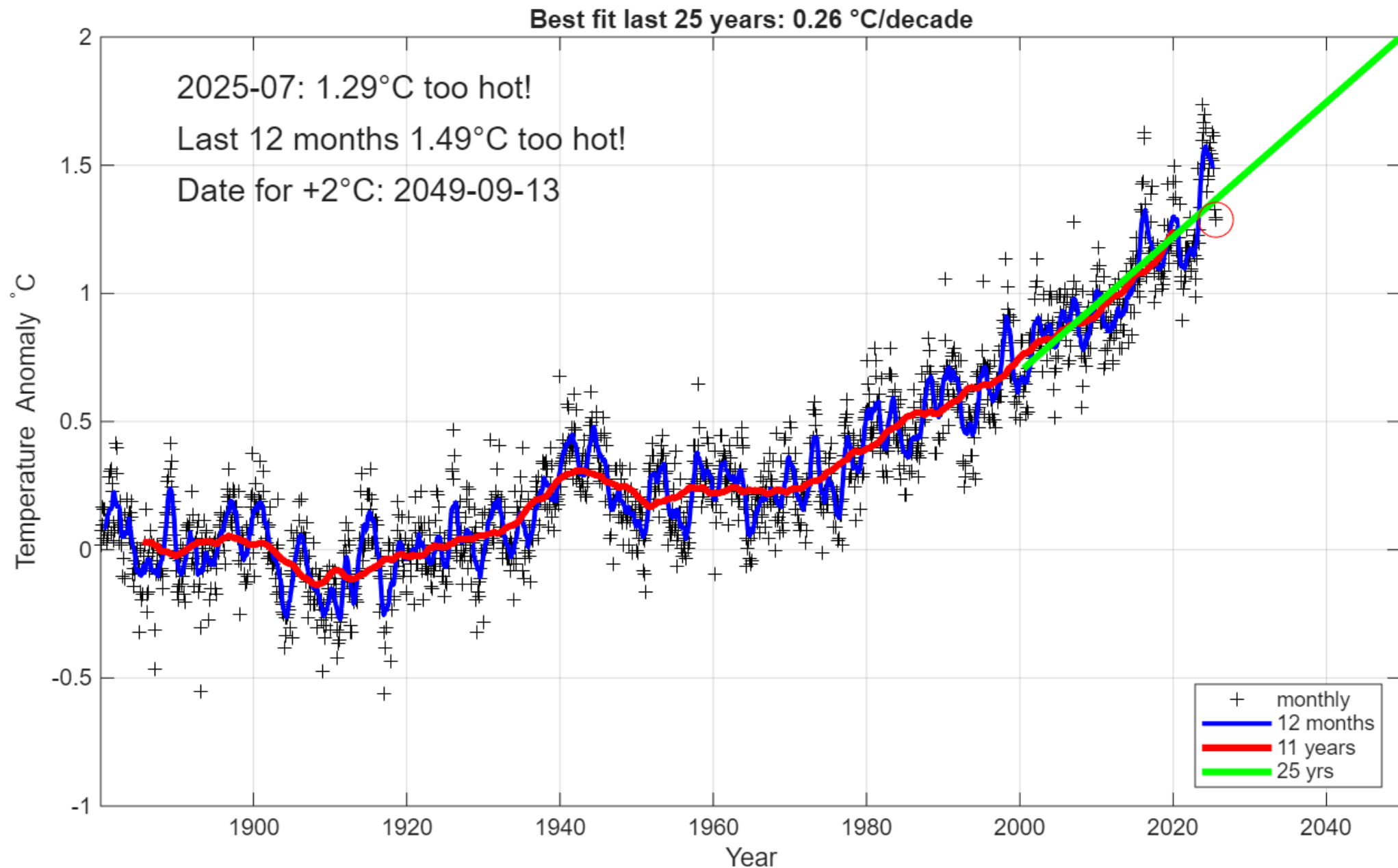
Earth's Energy Imbalance

Watts per square meter at 510,000,000,000 m²

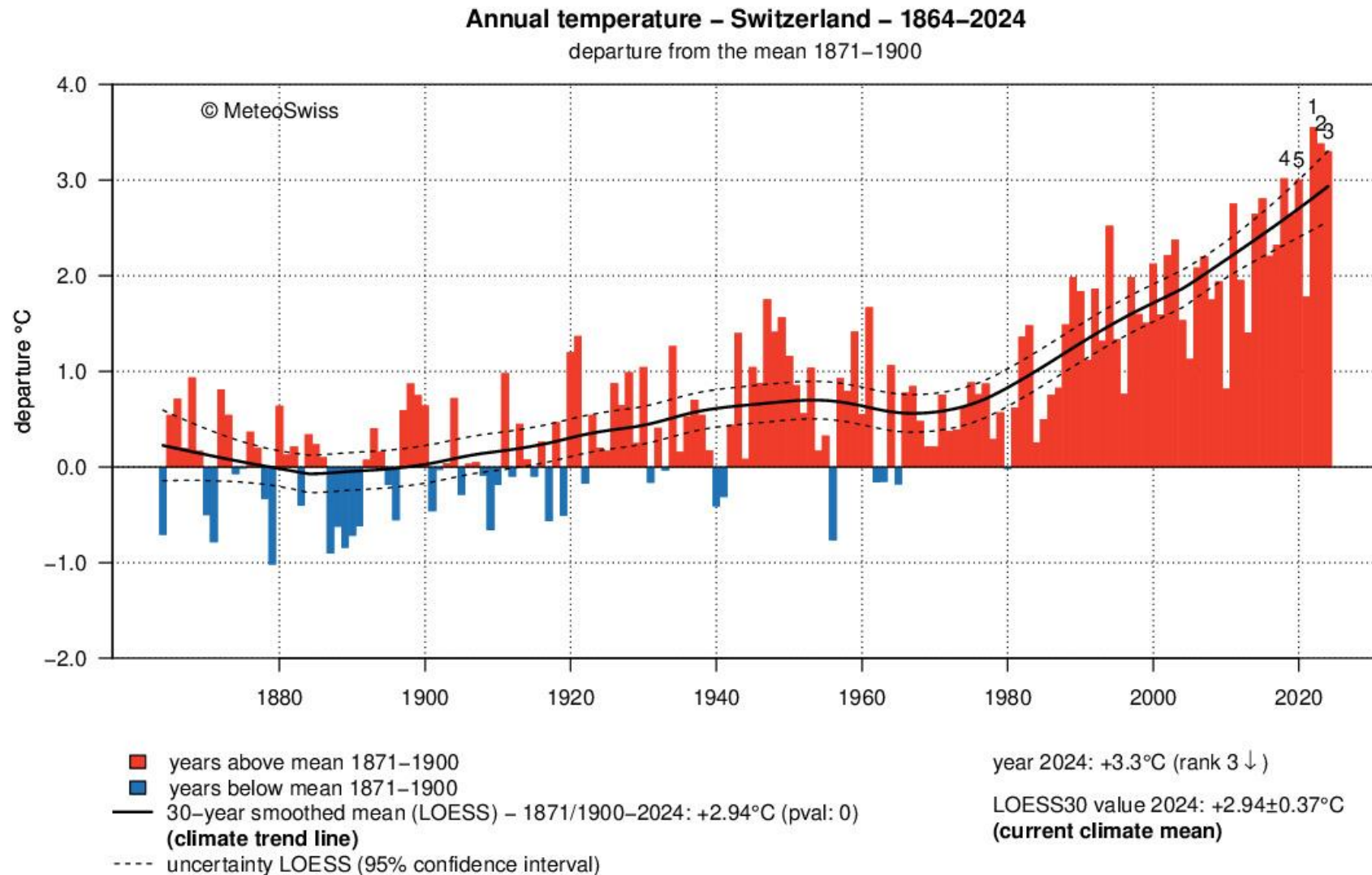


©Leon simons - Data source: Sarah Shackleton et al., Nature Geoscience (2023); von Schuckmann (2023); NASA CERES EBAF-TOA All-sky Ed4.2 Net flux

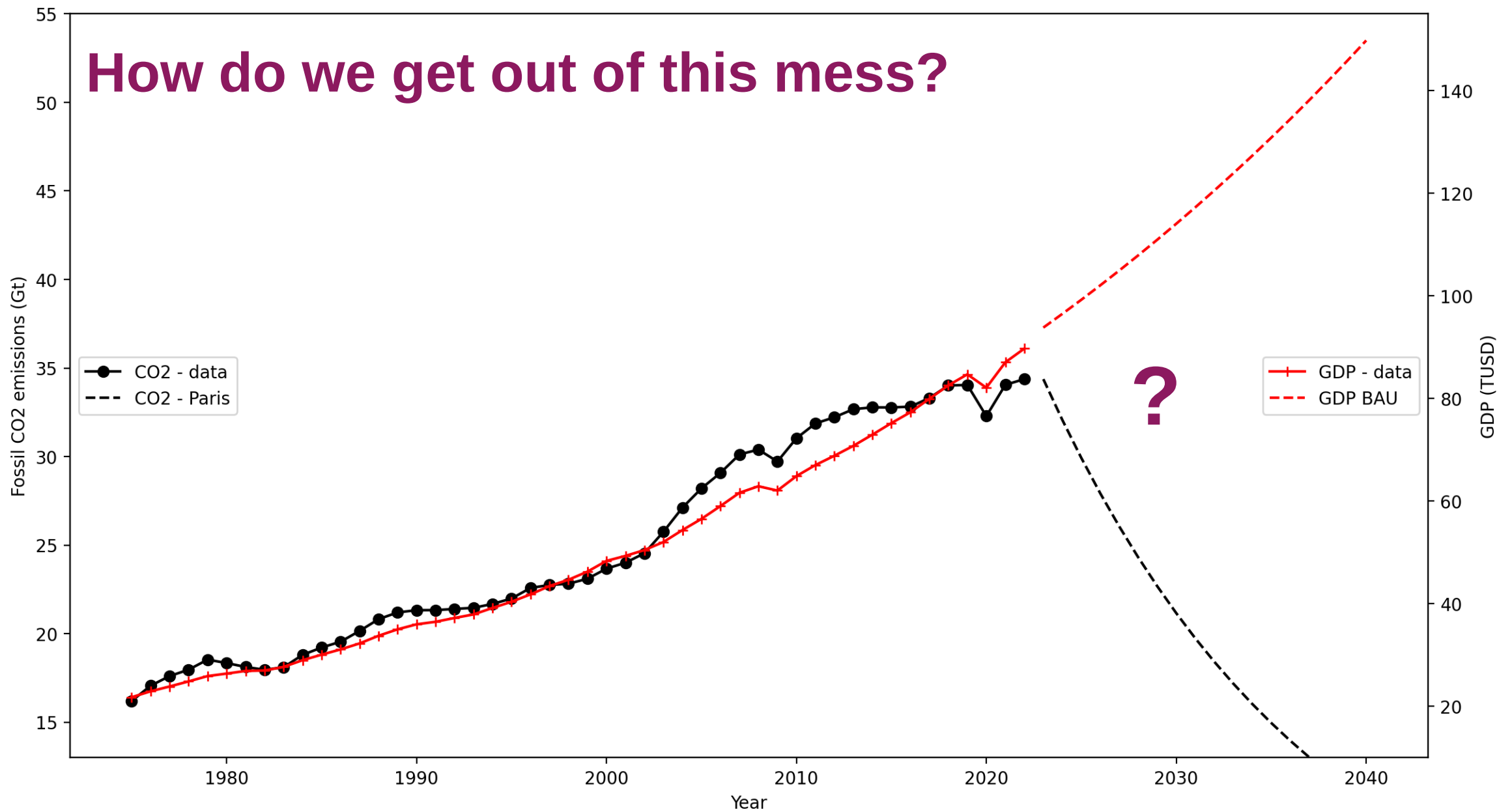




Average Temperature in Switzerland

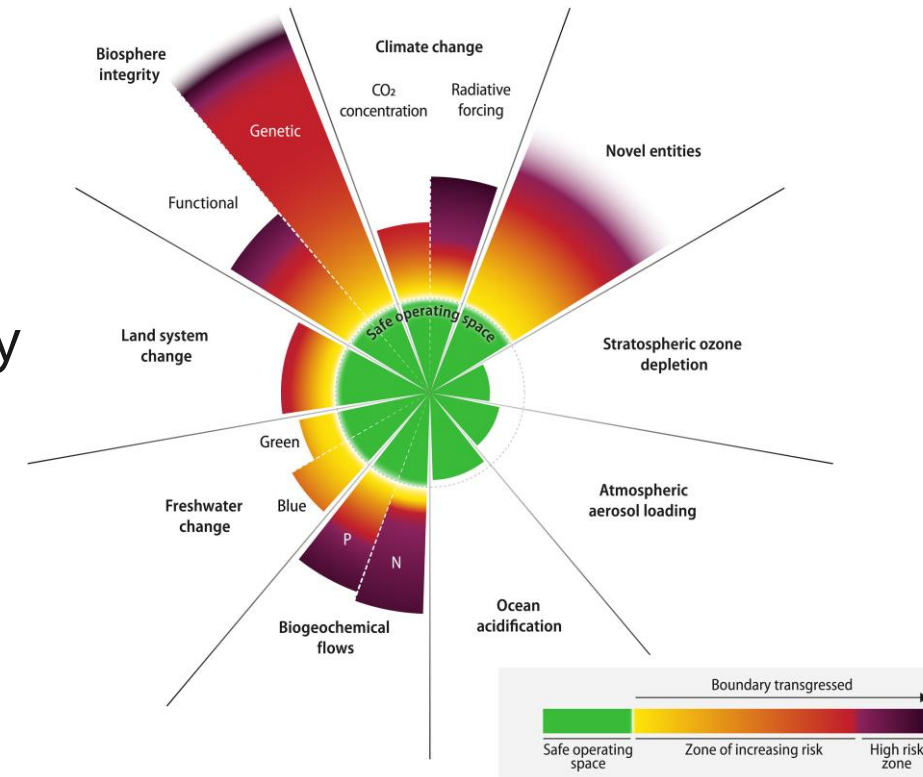


How do we get out of this mess?



Ignoring reality out of political convenience

- Anyone who believes in indefinite growth in anything physical, on a physically finite planet, is either mad or an economist.
– Kenneth E. Boulding
- Wackernagel et al., Global Footprint Network (Globally 1.75, Switzerland 4.4)
(<https://www.footprintnetwork.org>)
- Rockström et al., Earthbeyond six of nine planetary boundaries, <https://doi.org/10.1126/sciadv.adh2458>
- Every single scientific assessment concludes that we are consuming too much!!

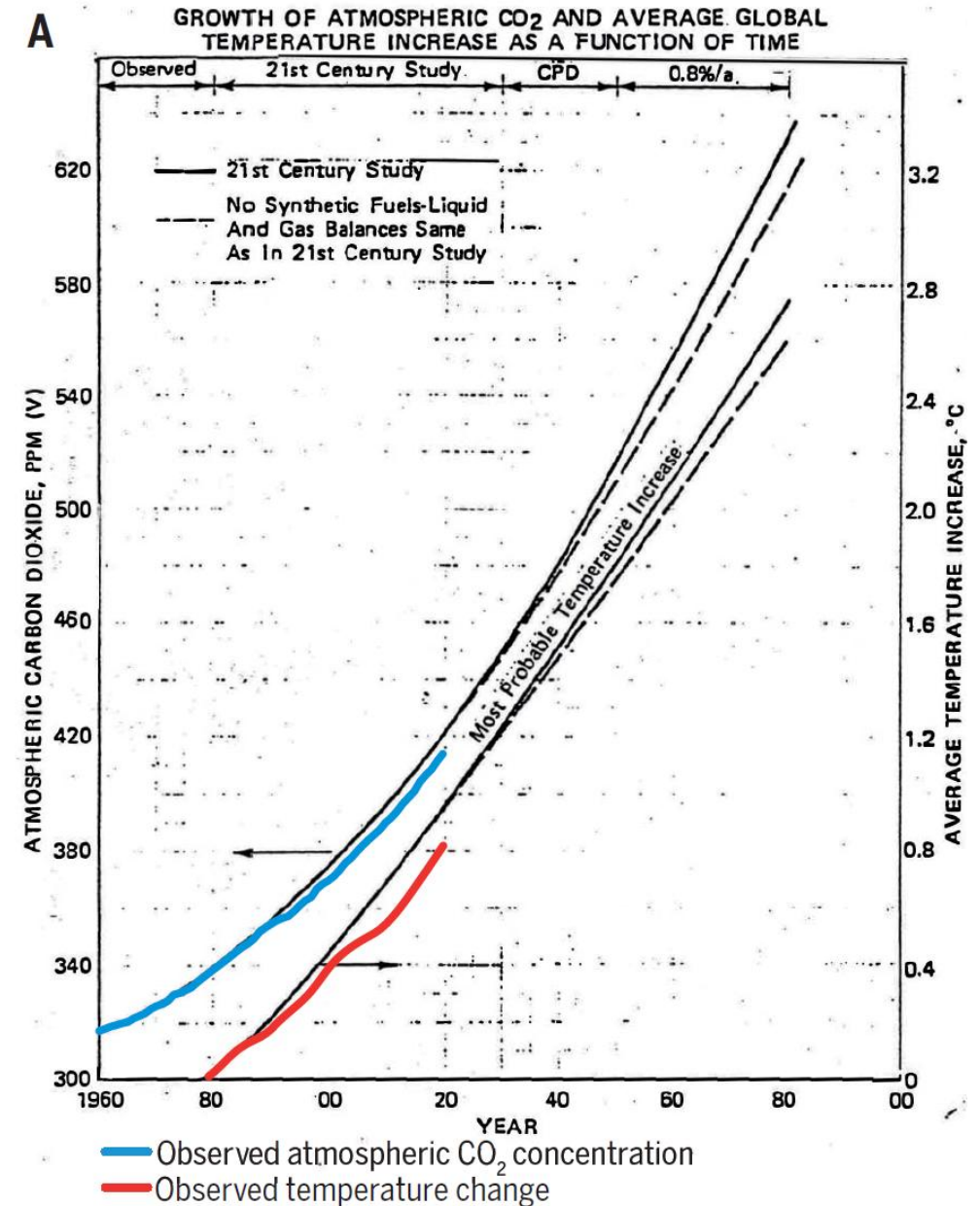




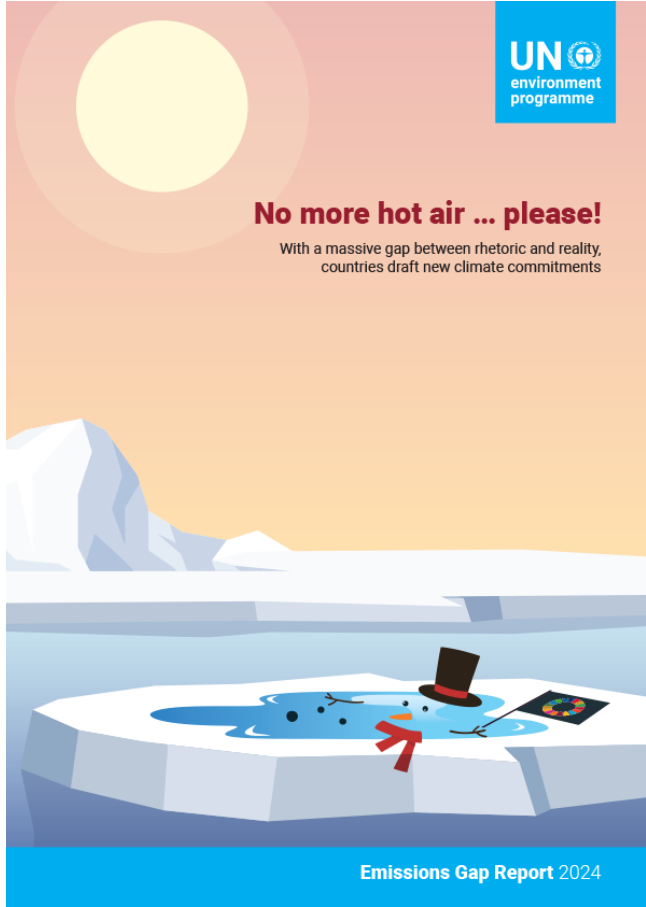
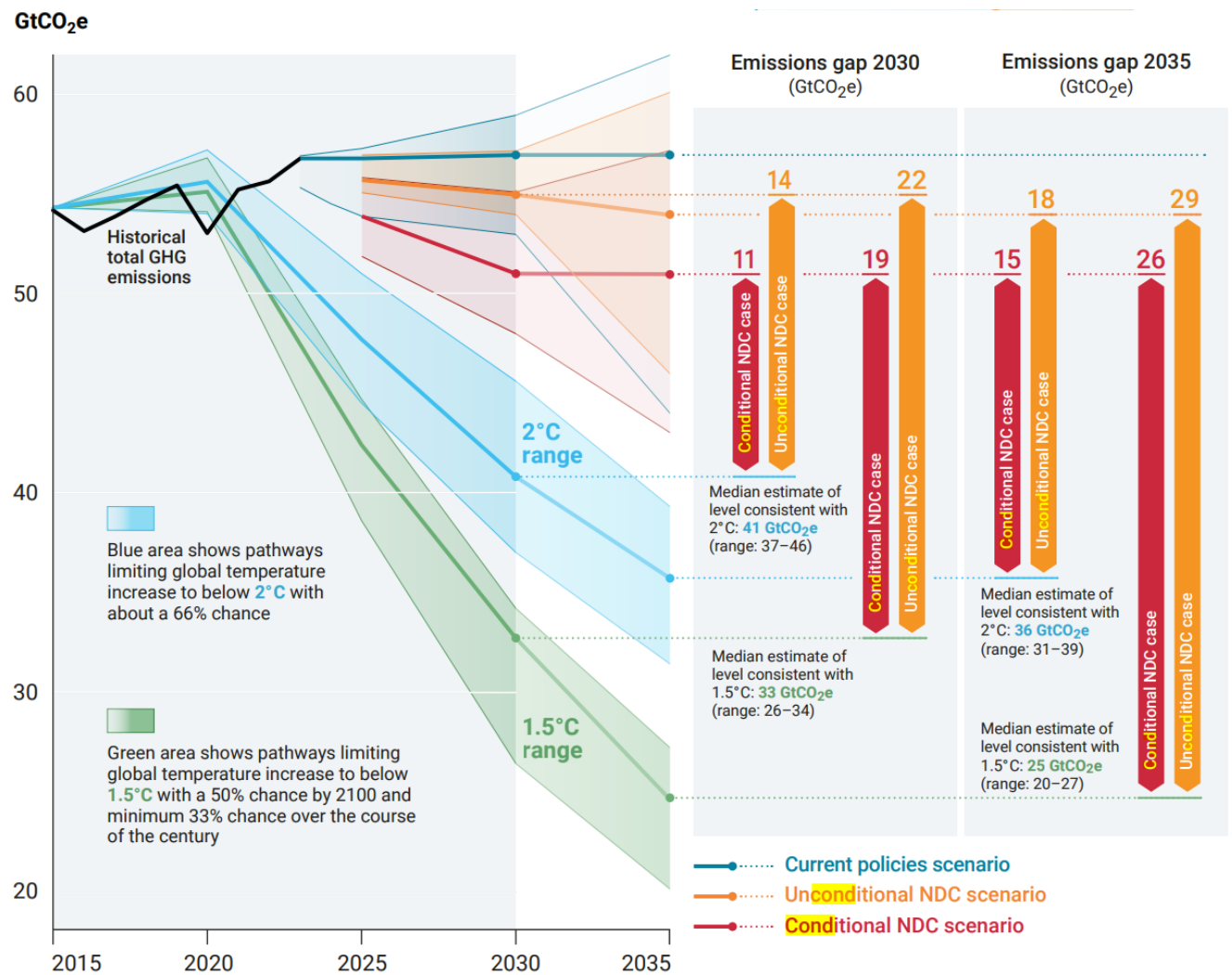
One cannot eat more and sh*t less!

Big Oil always new

- G. Supran*, S. Rahmstorf, N. Oreskes, Assessing ExxonMobil's global warming projections, DOI: 10.1126/science.abk0063
- 1982 Memo to Exxon Management about CO2 Greenhouse Effect, <https://www.climatefiles.com/exxonmobil/1982-memo-to-exxon-management-about-co2-greenhouse-effect>
- *It is difficult to get a man to understand something, when his salary depends on his not understanding it.* – Upton Sinclair

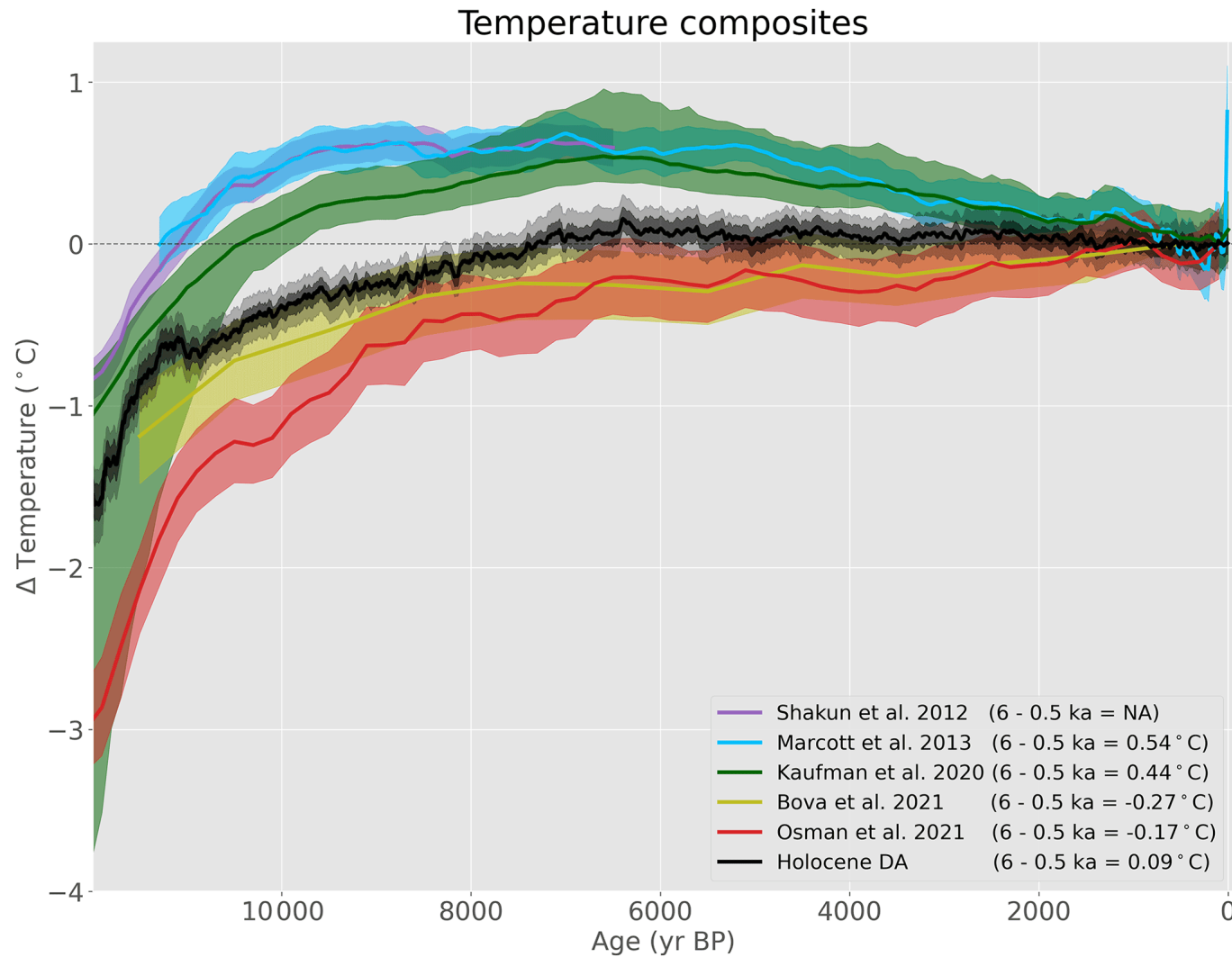


No more hot air ... please!



*Emissions Gap Report 2024:
No more hot air ... please!*

Temperature of the holocene



Moosehead of stone, Kuusamo, Finland
(approx. 4'000 years old)



Peasant with a plow, Egypt,
(approx. 4'000 years old)

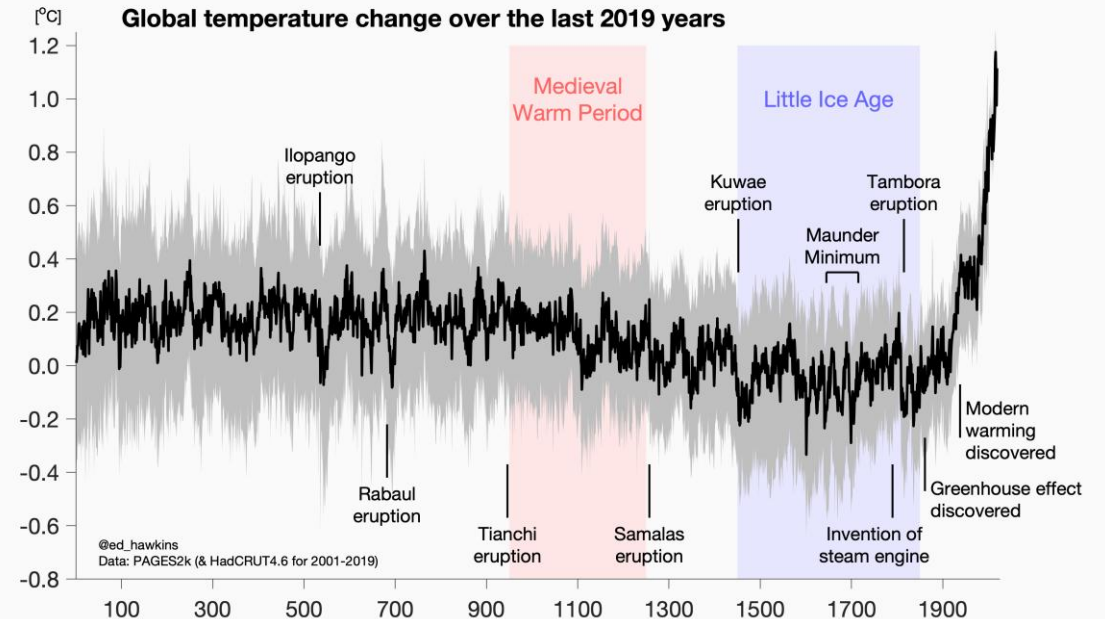
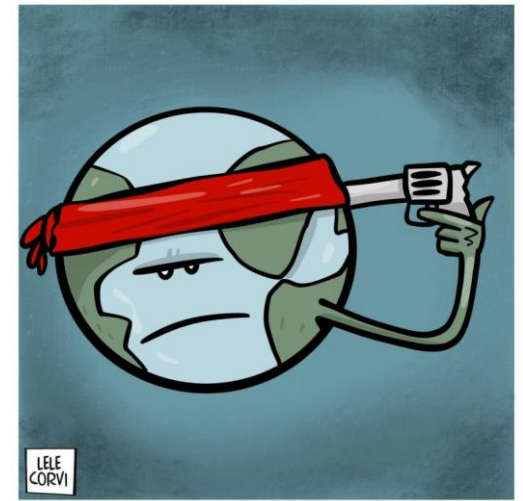
No, the climate has not always changed!

- Achilles' spear was made of ash ...
- as was the spear of Beowulf
- ... and the world ash tree (Yggdrasil)
- There are many very old trees:
 - The thousand-year-old oak, Bosjökloster, Sweden
 - Oliveira do Mouchao, Portugal (3,350 years old)
 - A farmer in the Middle Ages could predict the climate much better than today's climate researchers.



Russian roulette with the earth?

- Our planet is a very complex non-linear system.
- We know a stable operating point of the system: the Holocene.
- This operating point has made advanced civilizations possible; the history of humanity has taken place over the last 8,000 years.
- We are now leaving this operating point, fully aware that
 - other operating points are probably not compatible with human life.
 - there is no way back (irreversible changes).
 - we do not have a second planet.



The Arctic: climate change's great economic opportunity

An enormous prize is on offer. When might it be grasped?



For bears of both the market and polar kind, a planet without an ice cap is a tragedy. The Arctic is warming four times faster than the world at large, causing ice to shrink by an area the size of Austria each year. Since the 1980s the volume of ice has fallen by 70% or more. The Arctic's first ice-free day may occur before 2030.

The Economist, January 23, 2025

CRITICAL REVIEW

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Carbon dioxide as a pollutant: the risks on human health and the stability of the biosphere

Ugo Bardi,^a Phil Bierwirth,^b Kuo-Wei Huang^c and John McIntyre^d

The consequences of the human-caused increase in carbon dioxide concentration in the atmosphere are normally discussed mainly in terms of its radiative forcing effect and the consequent global warming and climate change. However, CO₂ is a chemically active molecule in aqueous environments, and it has multiple effects on the biosphere. Increasing CO₂ concentrations in the atmosphere increase the acidity of seawater and harm marine organisms. High CO₂ concentrations can make the photosynthetic reaction faster in some plants but also negatively affect the metabolism of aerobic metazoans, with harmful effects on human health. These effects are already important for people living in closed spaces and can only become stronger with the projected future increases in CO₂ atmospheric concentration. The present paper is a critical review of a field that is important for the future of humankind. We find that the chemical and biochemical pollution associated with CO₂ is a serious problem that may turn out to

Direct effects of CO₂ poisoning.

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ENVIRONMENTAL RESEARCH LETTERS

LETTER

Shutdown of northern Atlantic overturning after 2100 following deep mixing collapse in CMIP6 projections

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Supplementary material for this article is available [online](#)



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Abstract

When will we pull the emergency brake?

The New York Times

<https://www.nytimes.com/2024/11/13/opinion/climate-change-heat-planet.html>

OPINION
GUEST ESSAY

We Study Climate Change. We Can't Explain What We're Seeing.

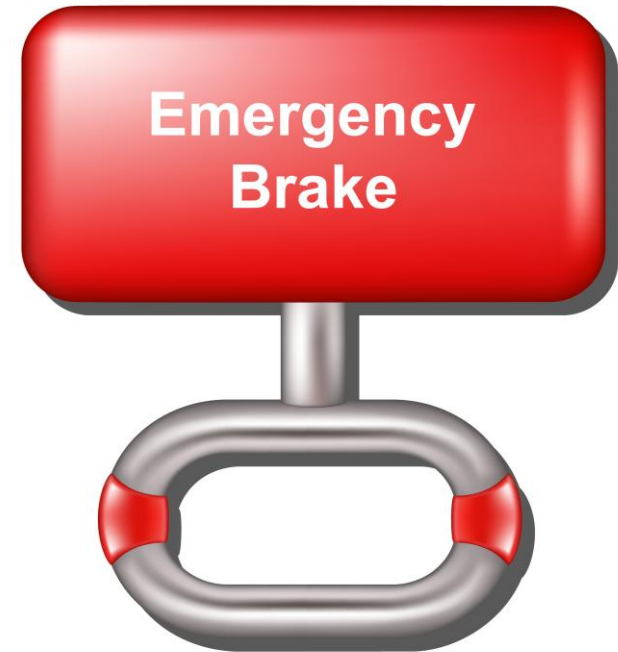
By Gavin Schmidt and Zeke Hausfather

Dr. Schmidt is a climate scientist in New York City. Dr. Hausfather is the climate research lead at Stripe and a

To fix this, we need to create a better way for climate models to reflect new observations.

warming below 1.5 degrees Celsius over preindustrial levels.

We know human activities are largely responsible for the long-term temperature increases, as well as sea level rise, increases in extreme rainfall and other consequences of a rapidly changing climate. Yet the unusual jump in global temperatures starting in mid-2023 appears to be higher than our models predicted (even as they generally remain within the expected range).





Why did we fail?

Die Klimakrise: Stehen wir am Abgrund oder sind wir schon einen Schritt weiter?

Henrik Nordborg

2025-09-08

Cutting-edge technology from the Bronze Age



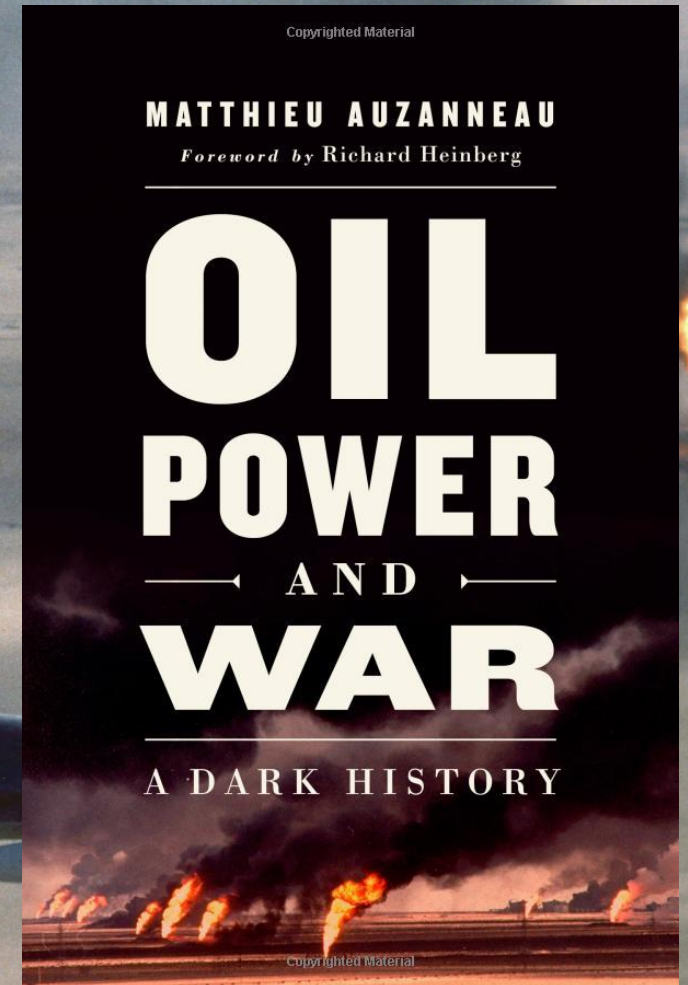
Cutting-edge technology from the Iron Age



The Fossil Hydrocarbon Age

If you depend on oil,
you need an army.

If you have an army,
you depend on oil.



Without peace there is no hope!

So, let us not be blind to our differences – but let us also direct attention to our common interests and to the means by which those differences can be resolved. And if we cannot end now our differences, at least we can help make the world safe for diversity. For, in the final analysis, our most basic common link is that we all inhabit this small planet. We all breathe the same air. We all cherish our children's future. And we are all mortal.

John F. Kennedy

Murdered
November 22, 1963

I refuse to accept the view that mankind is so tragically bound to the starless midnight of racism and war that the bright daybreak of peace and brotherhood can never become a reality. I refuse to accept the cynical notion that nation after nation must spiral down a militaristic stairway into the hell of thermonuclear destruction.

Martin Luther King

Murdered
April 4, 1968

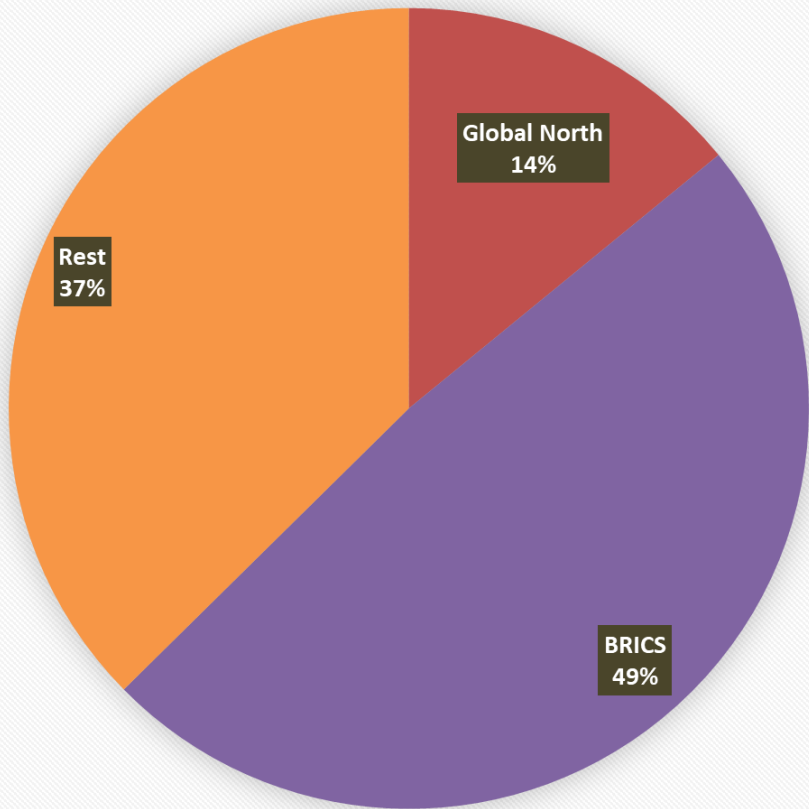
I don't want to be part of a government, I don't want to be part of the United States, I don't want to be part of the American people and have them write of us as they wrote of Rome: "They made a desert, and they called it peace."

Robert F. Kennedy

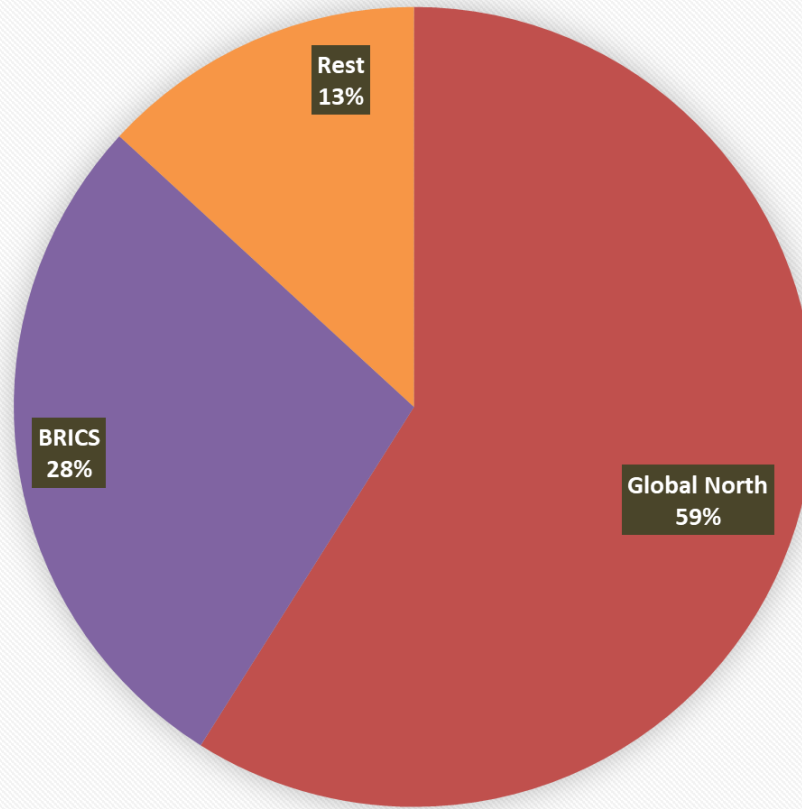
Murdered
June 6, 1968

Our unequal world

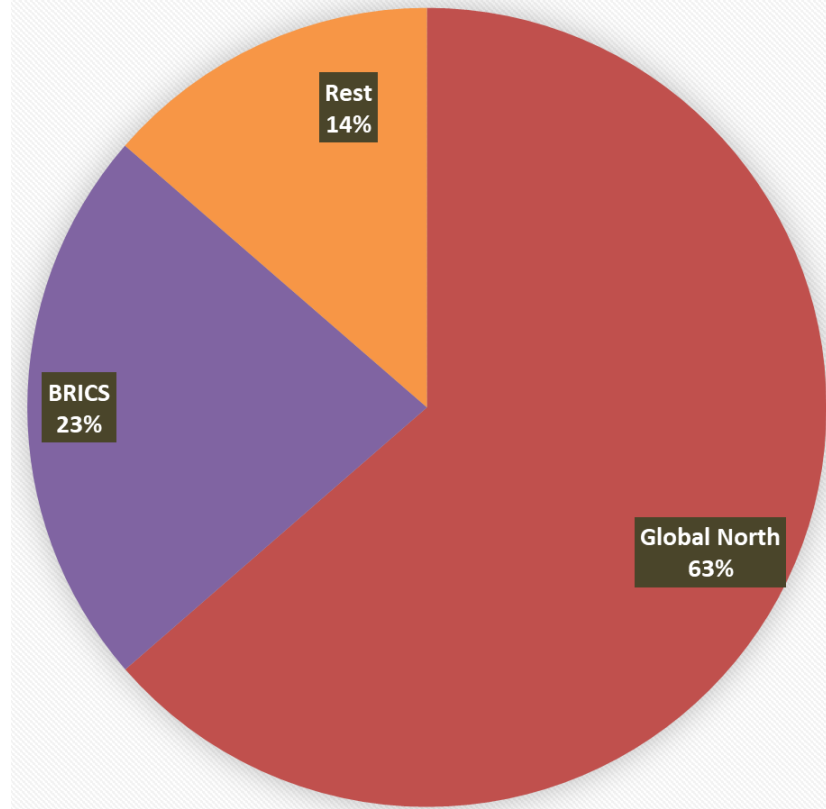
Population



Gross Domestic Product (GDP)



Military Expenses



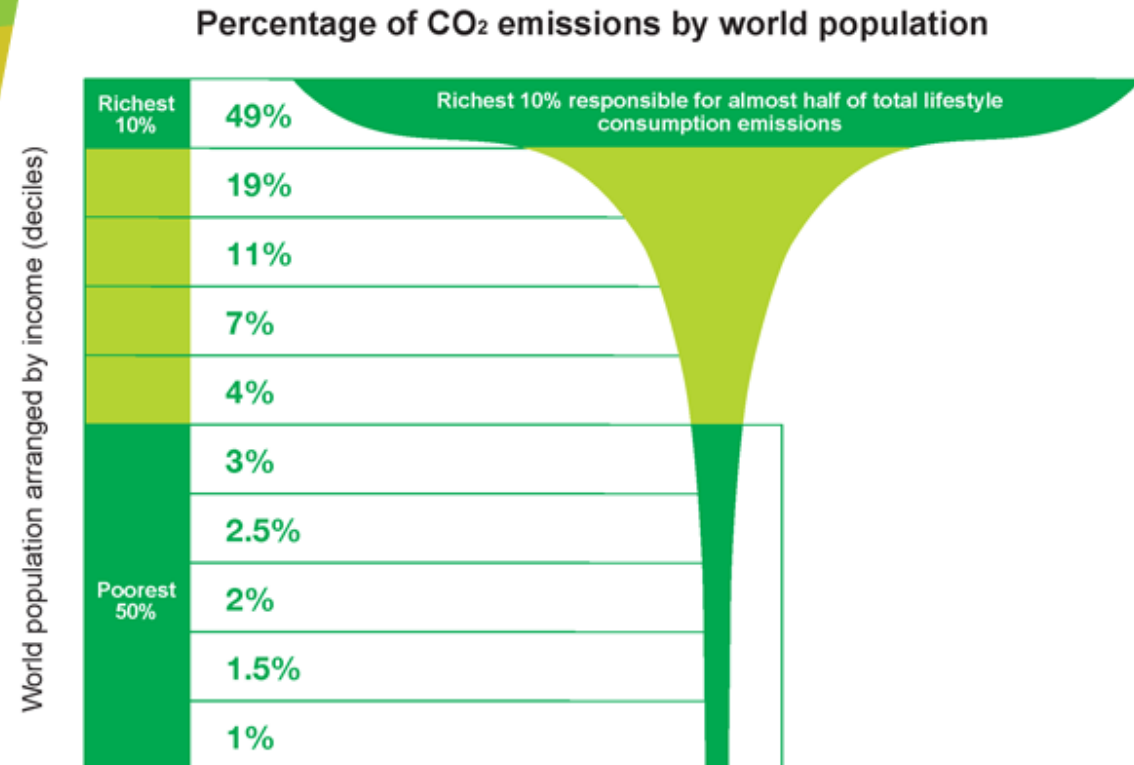
Climate Injustice: The Elephant in the Room



OXFAM

The poorest **50%** is only responsible for around **10%** of total lifestyle consumption emissions.

OXFAM MEDIA BRIEFING:
EXTREME CARBON INEQUALITY



Don't take my word for it ...

We have about 50% of the world's wealth but only 6.3% of its population. This disparity is particularly great as between ourselves and the peoples of Asia. In this situation, we cannot fail to be the object of envy and resentment. Our real task in the coming period is to devise a pattern of relationships which will permit us to maintain this position of disparity without positive detriment to our national security. **To do so, we will have to dispense with all sentimentality and day-dreaming; and our attention will have to be concentrated everywhere on our immediate national objectives.**

...

We should cease to talk about vague and -- for the Far East -- unreal objectives such as human rights, the raising of the living standards, and democratization. **The day is not far off when we are going to have to deal in straight power concepts. The less we are then hampered by idealistic slogans, the better.**

– George Kennan, 1948

The American way of life is not up for negotiations. Period.

– George H.W. Bush, 1992

Geopolitics is Simple

- The Global North is using far too many resources.
- It depends on the import (or extraction) of resources from the Global South.
- The US military (or NATO) exists to ensure access to these resources.



Crude Oil Reserves in Billion Barrels (Gbbl)



Note: For visualization purposes we are showing only countries with 100,000,000 bbl (0.1 Gbbl) of crude oil reserves or more.
How to read this map: Countries appear bigger as their crude oil reserves are bigger. e.g. Venezuela. Conversely, countries that have smaller reserves of crude oil appear smaller. e.g. Côte d'Ivoire
Article & Sources:
<https://howmuch.net/articles/worlds-biggest-crude-oil-reserves-by-country>
Central Intelligence Agency - <https://www.cia.gov/library>

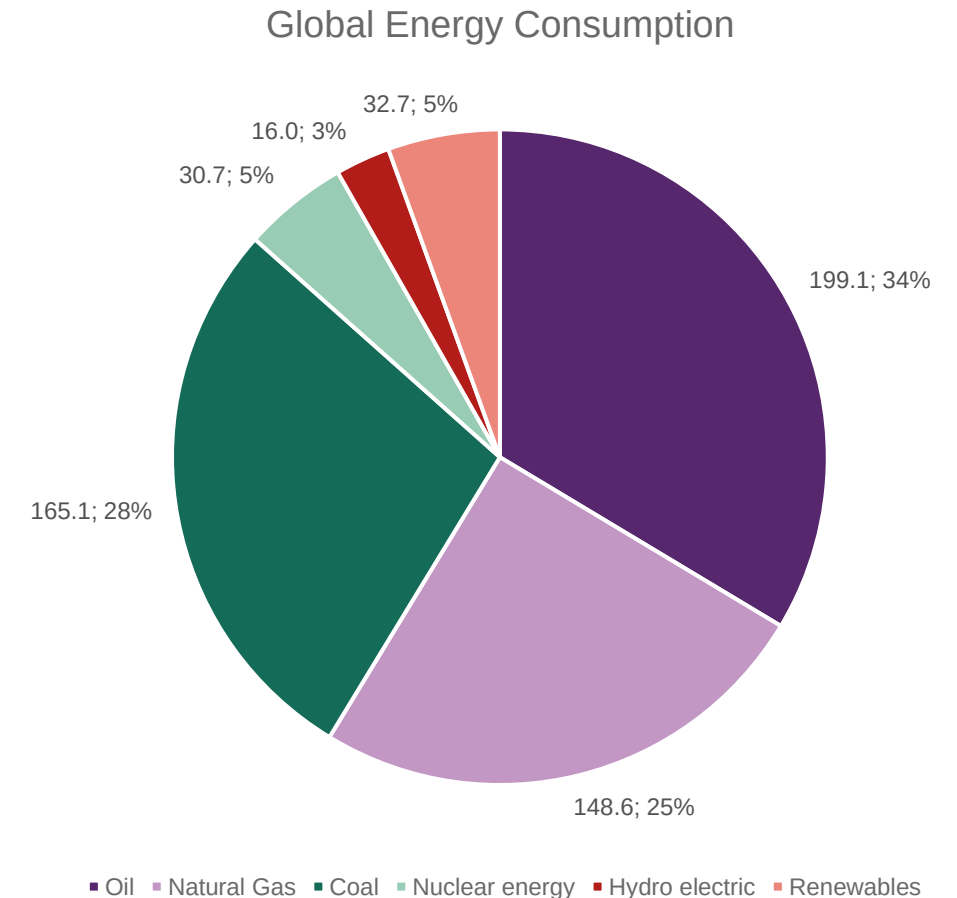
COP is a planned failure

The Assemblies of those early years were worth attending, for the Foreign Ministers of the Great Powers had not yet realised how easily, by means of a little tact and some elegant camouflage, the League might be used as a stage on which they could play the skilled game of the Old Diplomacy circumspectly dressed up in international costume. Before 1925, perhaps as many as fifty per cent of the delegates who went to Geneva honestly believed that the organisation of international peace was a workable proposition.

Vera Brittain, Testament of Youth

Competition is the root of all evil

- We live in the Fossil Hydrocarbon Age
 - More than 85% of global energy demand is still provided by fossil fuels.
 - Fossil hydrocarbons provide the raw material of the petrochemical industry.
 - All wars since the beginning of the 20th century have been powered by oil and most of them have been fought over oil.
- Our society will look completely different without fossil fuels.
 - Decarbonization will make countries poorer (in terms of GDP) and less powerful.
 - No economic growth = no capital income = no financial sector and no armies.
- The Ultimate Prisoner's Dilemma
 - Unilateral decarbonization is not possible.



The Devil's Excrement: One Fuel to Rule Them All

I call petroleum the devil's excrement. It brings trouble ... Look at this locura – waste, corruption, consumption, our public services falling apart. And debt, debt we shall have for years.

Juan Pablo Pérez Alfonso (1975)



An aerial photograph of a river with a complex, swirling pattern of dark blue and black water, indicating pollution. A large, rusted metal oil barrel is floating on the surface of the water. The barrel is tilted and has a small circular opening visible on its side. The water's surface is covered in intricate, vein-like patterns of dark and light blue, suggesting a mix of clean and polluted water.

Global Climate Compensation

Die Klimakrise: Stehen wir am Abgrund oder sind wir schon einen Schritt weiter?

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The Ultimate Prisoner's Dilemma

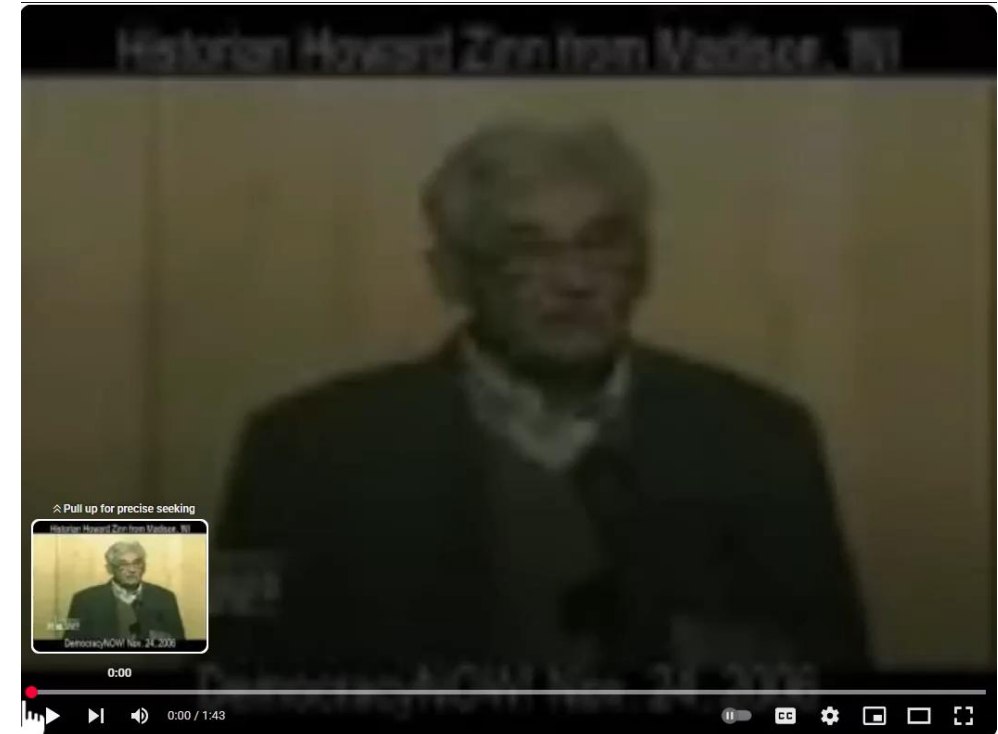


Climate Mitigation requires international collaboration.

Climate Adaption can be done unilaterally.

Governments Lie

- Howard Zinn:
 - Governments lie all the time. Since they do not represent the people that elected them, they simply must lie to stay in power.
- The US government is a case in point:
 - It lied about the Gulf of Tonkin incident at the outset of the Vietnam War.
 - Daniel Ellsberg: The Pentagon Papers.
 - WMDs in Iraq.
 -
- No country in the world has climate policy compatible with the Paris Agreement, but they are not ready to admit it.

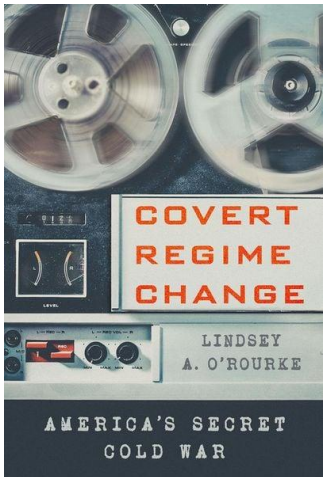
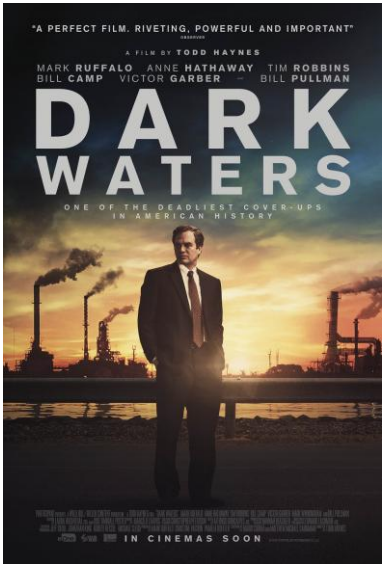
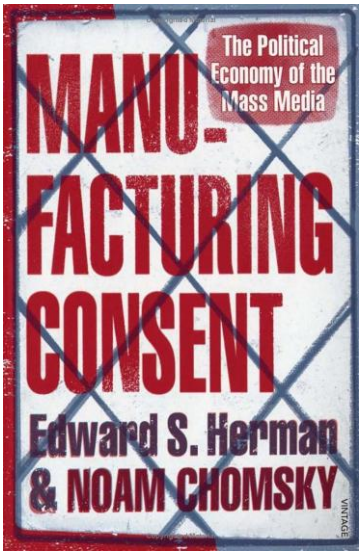
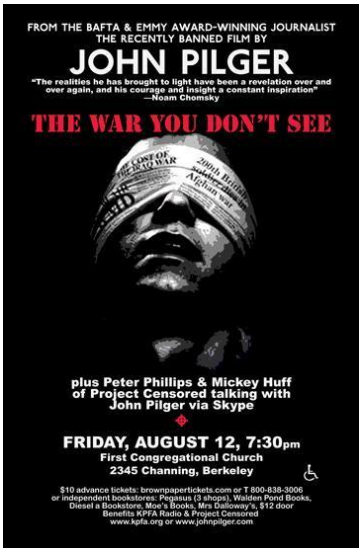


The Lies of Corporations – Greenwashing



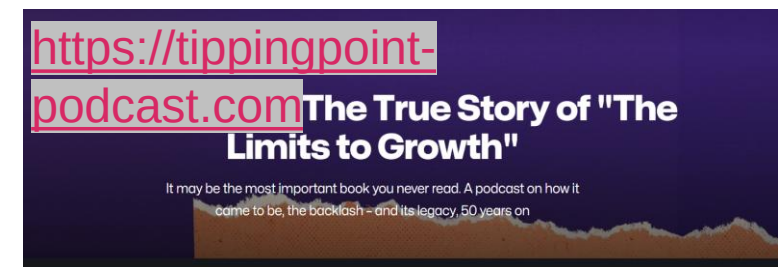
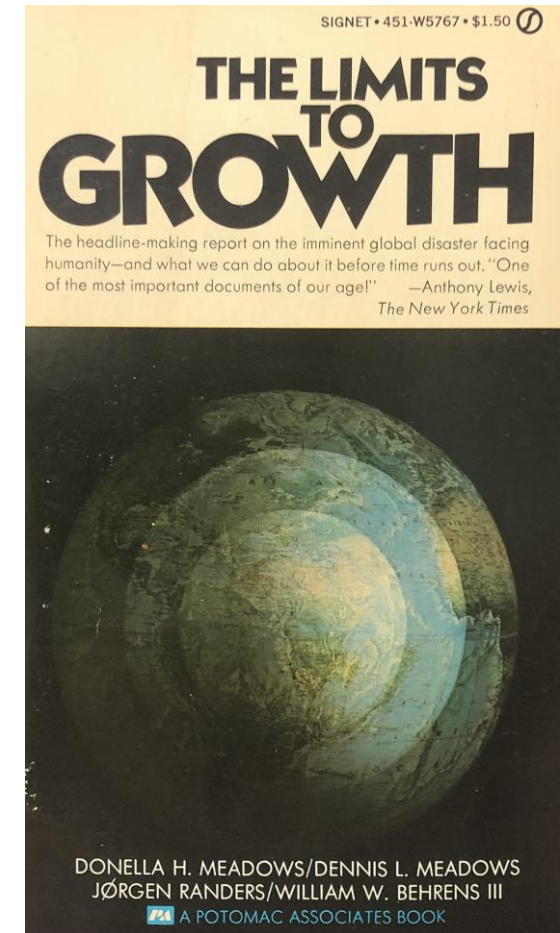
- Corporations are in it for the money.
- They regularly lie to improve their profits.
- Examples:
 - Tobacco Companies
 - Oil Companies
 - Drug Companies
 - Car Manufacturers
 -

Some examples ...

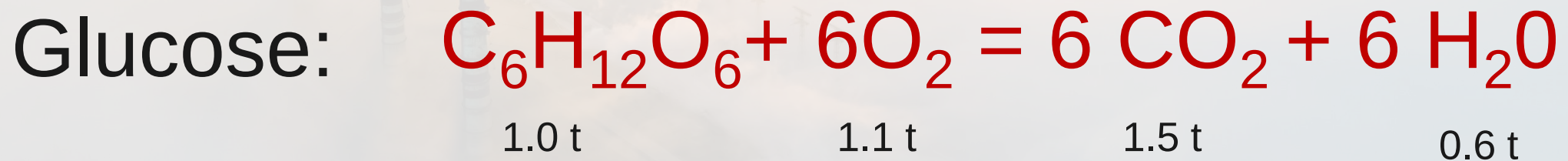
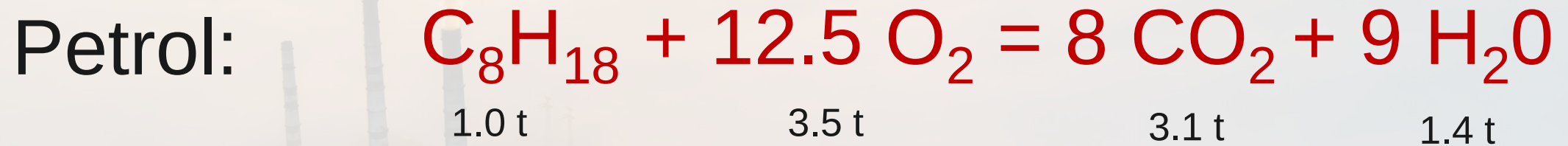
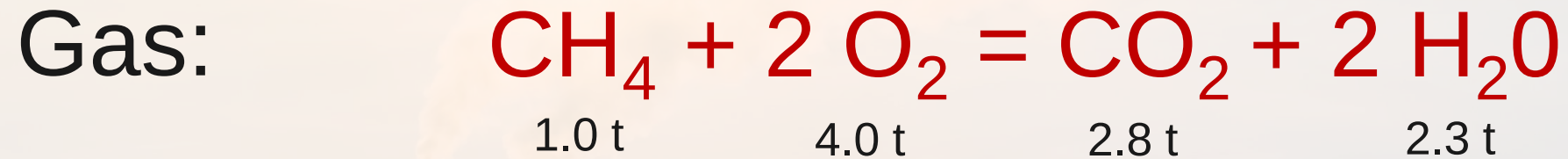


Limits to Growth (1972)

- Questioning the system was still not allowed in 1972.
- The usual suspects:
 - Economists realized that accepting the study meant the end of capitalism.
 - The fossil fuel industry wanted to continue with business as usual.
 - Engineers did not like the thought that technology could not solve all problems.
 - Politicians did not want to make unpopular decisions.
- Check the excellent podcast Tipping Point



Never run a combustion engine indoors!



A large open-pit mine with terraced levels and a worker in the foreground. The worker is wearing a white hard hat, a black shirt, and a black mask, and is holding a small device. The mine is a deep, multi-tiered excavation with visible rock layers and some vegetation on the upper slopes. The sky is blue with some clouds.

Do we have to destroy the planet to
save capitalism?

Let's agree to disagree.

Global Climate Compensation

1. All fossil fuel producers pay a fee proportional to their production to a global fund.
2. The money from the fund is distributed among the world's nations on a per capita basis.

We need only agree on the price of CO₂ emissions.



Quick Calculation

- Let us fix a price at 100 USD/tCO₂
 - The fund receives 3600 billion USD = 3.6 TUSD annually.
 - The price of a barrel of oil increases by 42 USD, or approximately 0.30 USD per liter of oil.
 - A global yearly income of 450 USD per capita or 38 USD per month to everyone.
 - Abolishes global poverty.
- Comparisons
 - Global GDP: 100 TUSD
 - Global defense spending: 2.2 TUSD
 - Global fossil fuel subsidies: 1.3 TUSD (or 7 TUSD)
 - Lost tax revenue due to tax havens: 0.5 TUSD.

Fossil Fuel Companies as Tax Collectors

- A very reasonable demand:
 - Easy to implement.
 - The corporations can pass on the costs to their customers.
- A global carbon tax:
 - Simple implementation (only a couple of hundred fossil fuel companies)
 - No loopholes, no tax havens.
 - Invalidates any business model based on fossil fuels.
 - Applies to the entire value chain.
- Appealing to governments:
 - Tax revenue without raising taxes.
 - Let the oil companies subsidize governments for a change.

Climate Justice Built In

- Goal: socialize all profits from fossil fuels.
- Most of the money from the fund would go to the Global South.
- At least 80% of the world's population would benefit.
- Paid out to governments with few strings attached.
- Complete Transparency.

The End of Greenwashing

- Any company that uses a lot of fossil fuels somewhere in the supply chain will have to pay.
- Companies that are carbon neutral will not see an increase in costs.
- If you are committed to “net zero”, you are either in favor of Global Climate Compensation or you are lying.

System Change – If Necessary

- Global Climate Compensation will significantly increase the price of fossil fuels.
- Two things can happen:
 - If the optimists are right, this will only speed up the transition to renewable energies, with a minimal cost to society.
 - If the pessimists are right, the economy will stop growing, meaning that we need to invent a new economic system (i.e. Degrowth).
- Either way, the planet benefits.

A Popperian Method to Save the World

- Nobody knows how fast we can transition away from fossil fuels.
- Targets like “Net Zero by 20XX” are worthless, because we do not know what is possible.
 - They can either be unrealistic or unambitious.
 - They allow policymakers to kick the can down the road.
- It is easier to predict the effects of a price increase.
- The price can be adjusted until we break the trend of rising greenhouse gas concentrations.

Moral Clarity

- The future of humanity rests in the hands of the oil companies.
- Fossil fuel companies are given a choice:
 - Either cooperate and pay Global Climate Compensation ...
 - ... or lose their social license to operate.
- Companies that do not cooperate will be destroyed:
 - Political campaigns
 - Legal action
 - Activism
 - Boycotts
 - ...
- Let us focus on one thing and get it done!



We are currently conducting an uncontrolled, irreversible, high-risk experiment with the only habitable planet in the known universe.

Why are we worried about conducting a controlled, reversible, risk-free experiment with the economic system?

"In the presence of a crime without a name"

A small group of very rich people are destroying the only habitable planet in the known universe.

They have the choice to stop doing so immediately.

It is our job to make sure that they want to do so.

"I wish it need not have happened in my time."

"So do I, and so do all who live to see such times. But that is not for them to decide. All we have to decide is what to do with the time that is given us."



The Once Vital Planet

As far as we know, only one planet in the universe has managed to outsmart the 2nd law of thermodynamics to create complex life.

That planet is now in the process of thermodynamic decay.



Our planet is dying.

Carbon dioxide and methane are created through the decay of organic matter.

They are the gases exuded by a rotting corpse.

Doing What is Right!

Hope has two beautiful daughters. Their names are anger and courage. Anger at the way things are, and courage to see that they do not remain the way they are.

St. Augustine

Once we cut through the lies, the greenwashing, and propaganda, what remains is the simple choice to do the right thing.

We should start tomorrow!